

PUBLISHED: 29-AUG-2017
2015.0 RANGE ROVER (LG), 303-01

ENGINE - TDV8 4.4L DIESEL

CRANKSHAFT REAR SEAL (G1509166)

REMOVAL AND INSTALLATION

12.21.20	OIL SEAL - CRANKSHAFT - REAR - RENEW	4400 CC, TDV8, WITH PARTICULATE FILTER, WITHOUT DIESEL EXHAUST FLUID	14.2	USED WITHINS	
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SPECIAL TOOL(S)

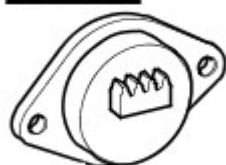


E130668

303-1237

Installer, Crankshaft
Position Sensor Ring

303-124



E81375

303-1243

Locking Tool,
Flexplate

PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME	
Installation Step 3	Renew Part	Rear crankshaft seal	1
Installation Step 9	Renew Part	Oil pan gasket	1
Installation Step 12	Renew Part	Retaining clip	1
Installation Step 15	Renew Part	Turbocharger oil return pipe gasket	1
Installation Step 15	Renew Part	Retaining clip	1
Installation Step 18	Renew Part	Flywheel bolts	8

REMOVAL



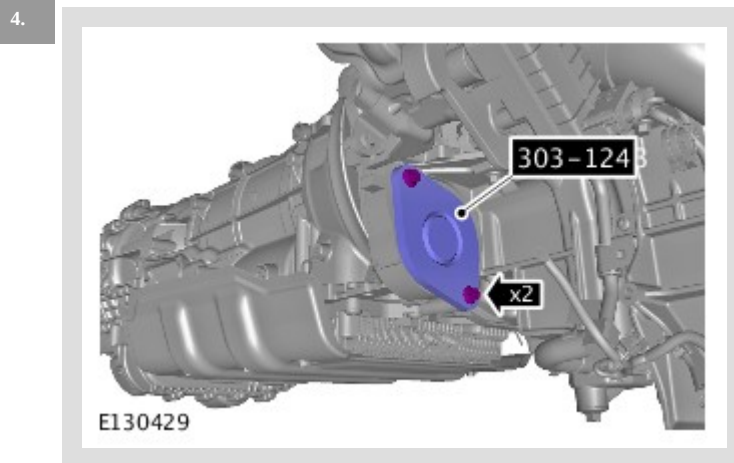
NOTES:

- Some variation in the illustrations may occur, but the essential information is always correct.
- Removal steps in this procedure may contain installation details.
- Some components shown removed for clarity.

1. Refer to: [Engine Oil Draining and Filling](#) (303-01E Engine - TDV8 4.4L Diesel, General Procedures).

2. Refer to: [Transmission - TDV8 4.4L Diesel](#) (307-01B Automatic Transmission /Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal).

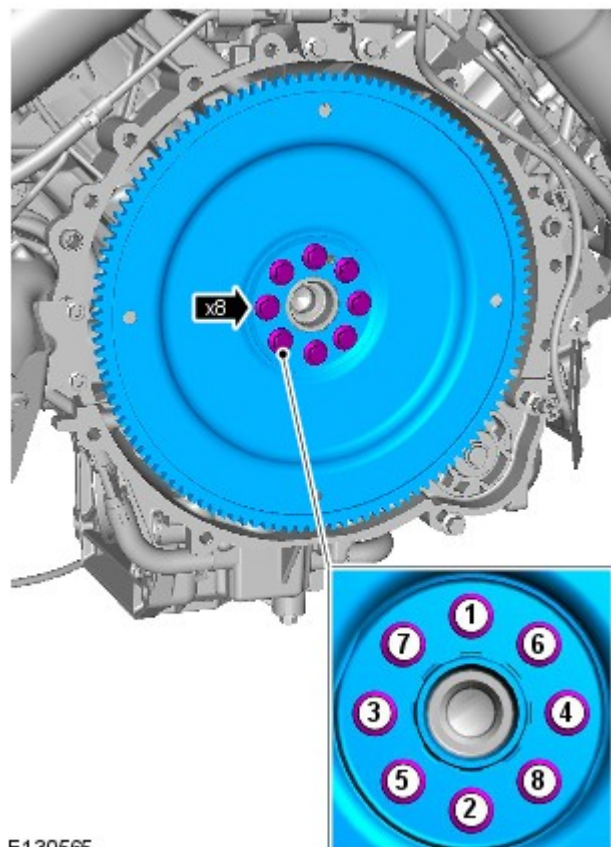
3. Refer to: [Differential Case](#) (205-03 Front Drive Axle/Differential, Removal and Installation).



Special Tool(s): [303-1243](#)

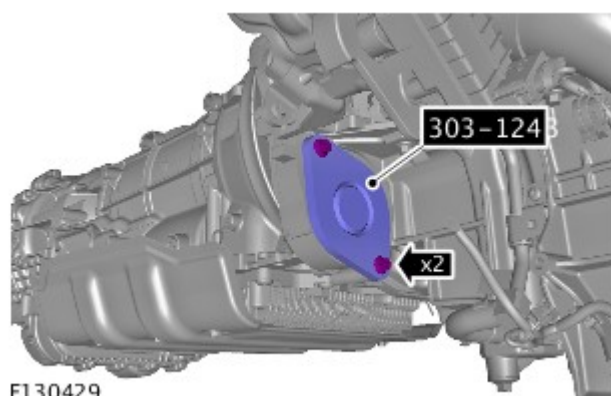
5.  **CAUTION:**

Discard the bolts.



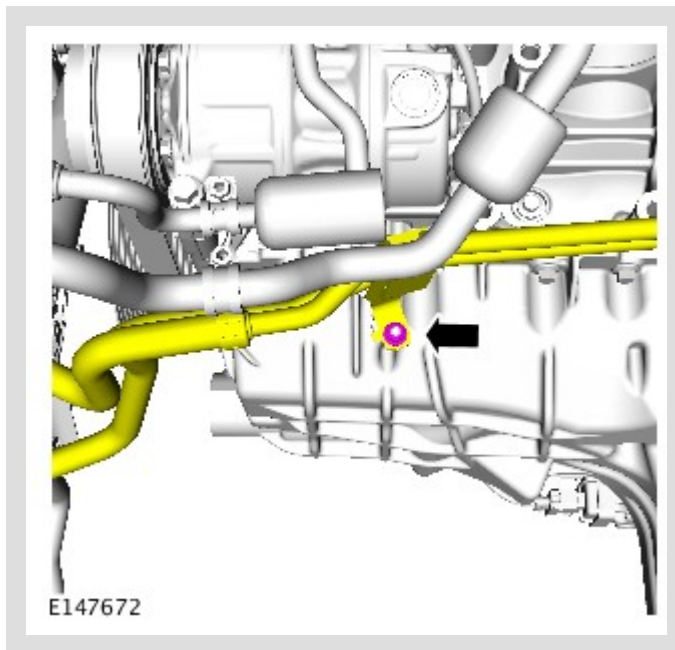
E130565

6.



E130429

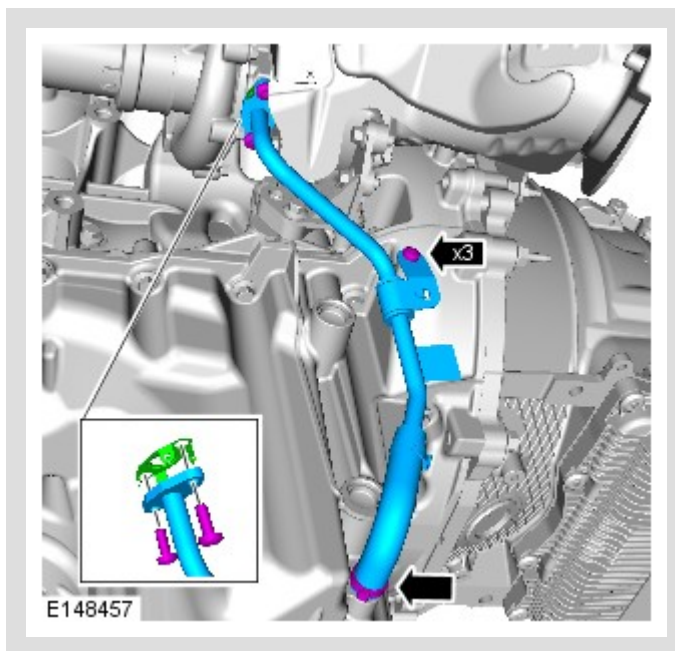
Special Tool(s): [303-1243](#)

**CAUTIONS:**

- Be prepared to collect escaping oil.
- Before disconnecting any components, make sure the area is clean and free from foreign material. When disconnected all openings must be sealed.

**NOTES:**

- Remove and discard the gasket.
- Discard the retaining clip.



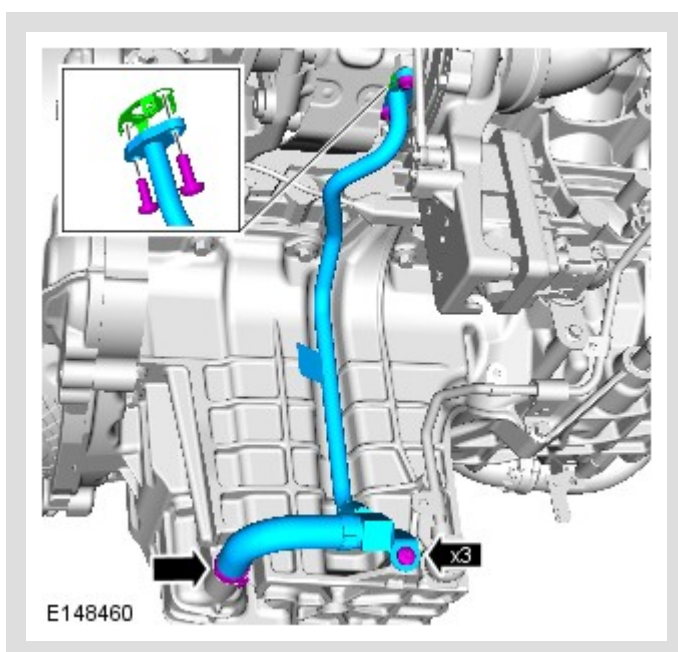
9.

⚠ CAUTIONS:

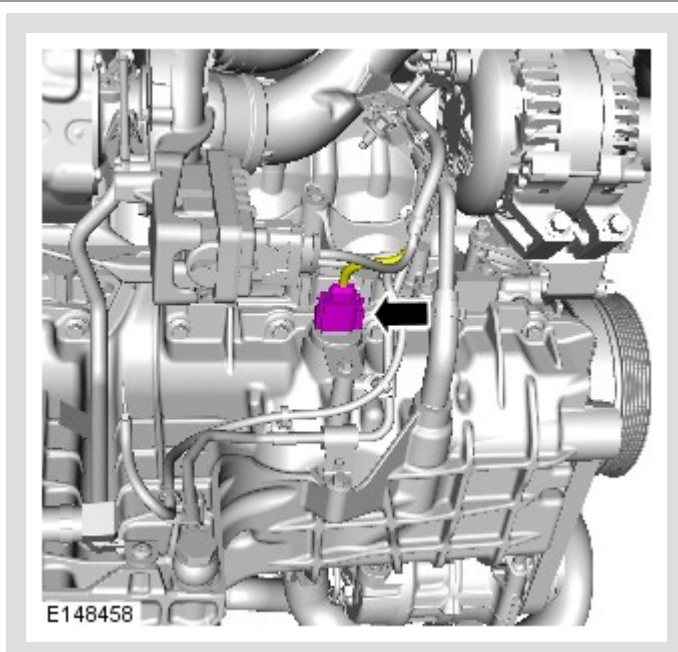
- Be prepared to collect escaping oil.
- Before disconnecting any components, make sure the area is clean and free from foreign material. When disconnected all openings must be sealed.

⚠ NOTES:

- Remove and discard the gasket.
- Discard the retaining clip.



10.



11.

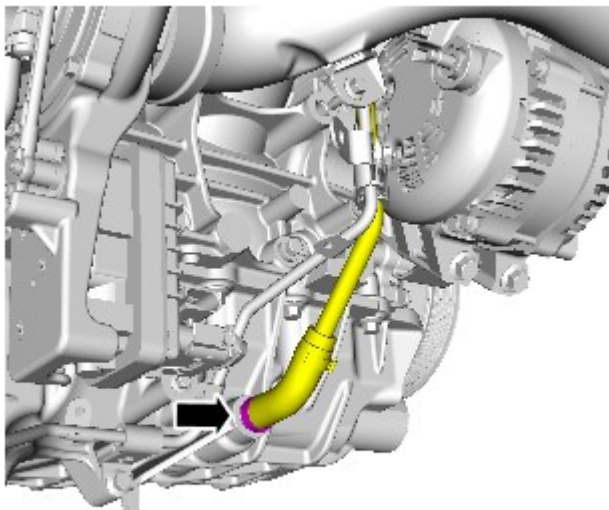
⚠ CAUTION:

Be prepared to collect escaping oil.



NOTE:

Discard the retaining clip.



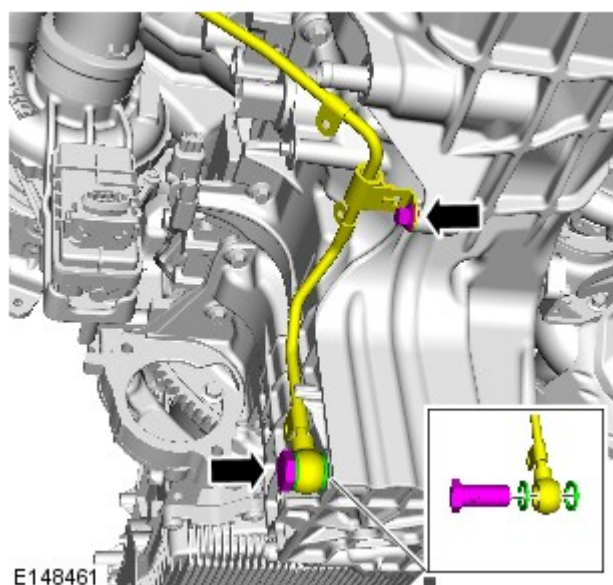
E148459

12.

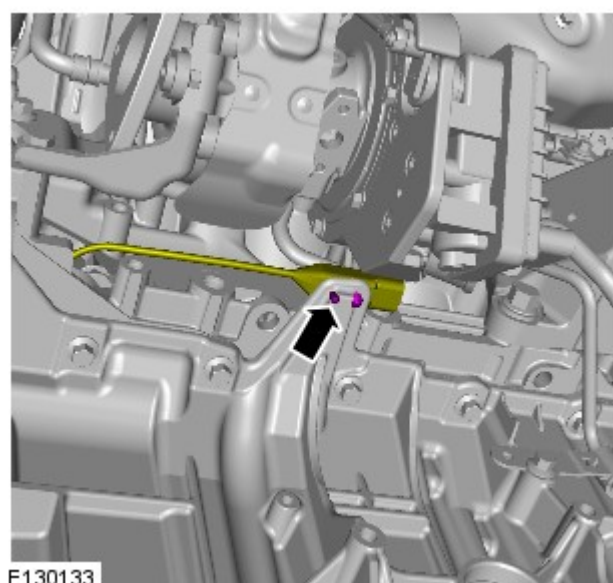


CAUTIONS:

- Be prepared to collect escaping oil.
- Inspect the seals, replace if damaged.



13.

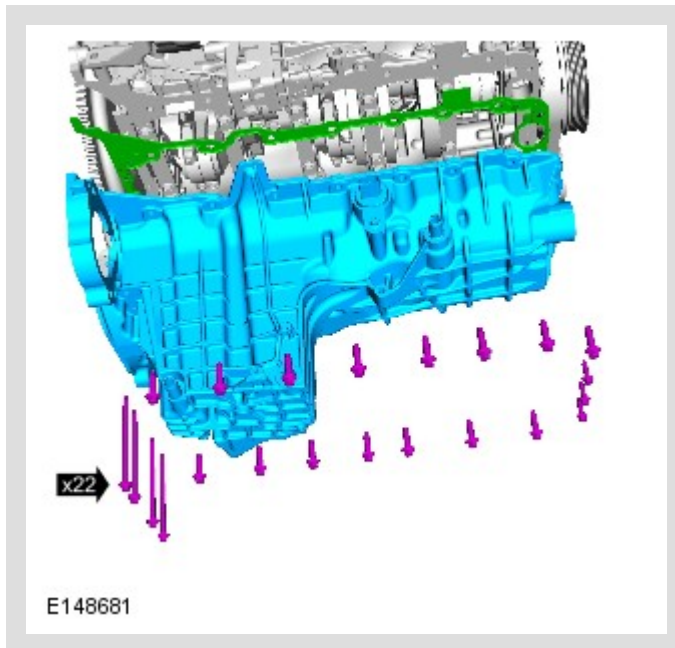


14.



NOTE:

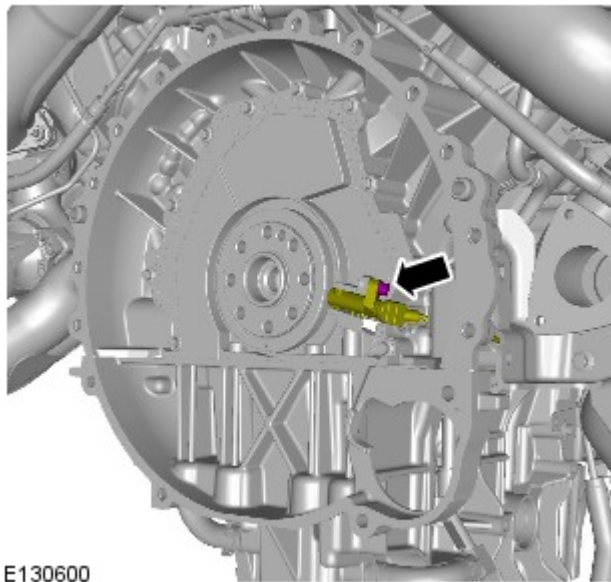
Remove and discard the gasket.



15.

⚠ CAUTION:

The CKP sensor bolt should not be removed from the CKP sensor.

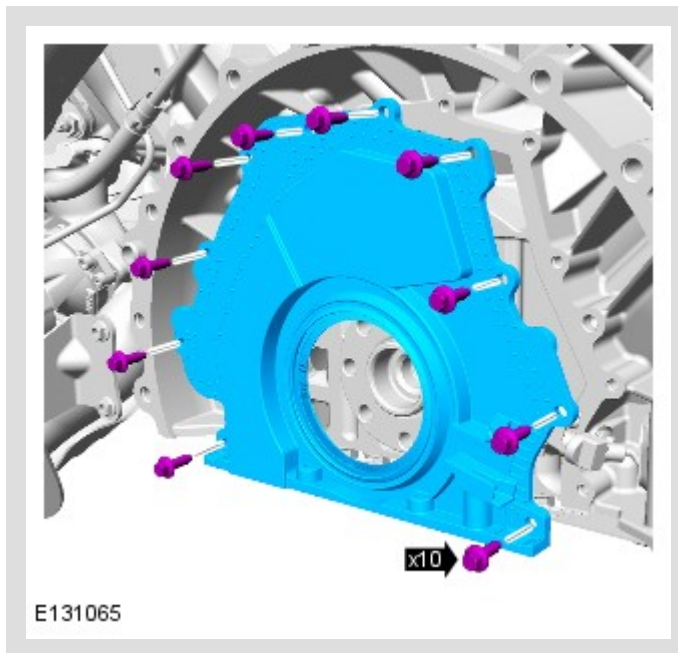


16.

⚠ CAUTION:

Discard the component.



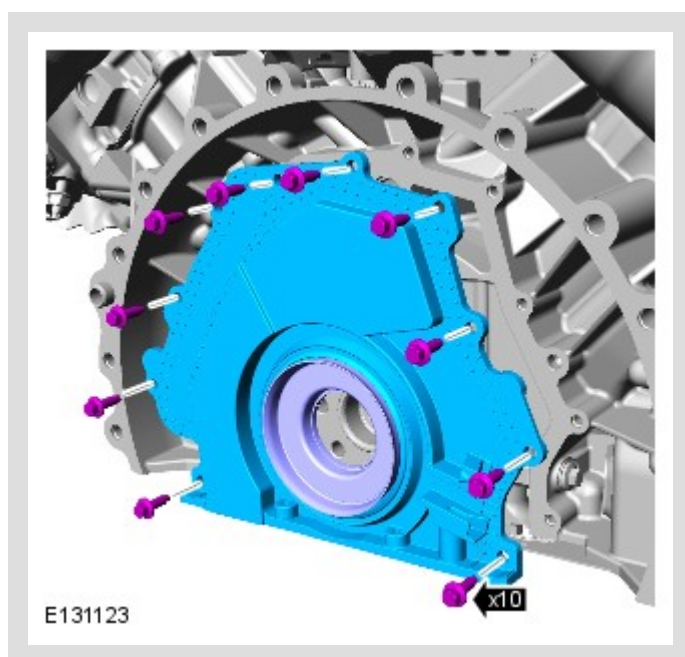


INSTALLATION

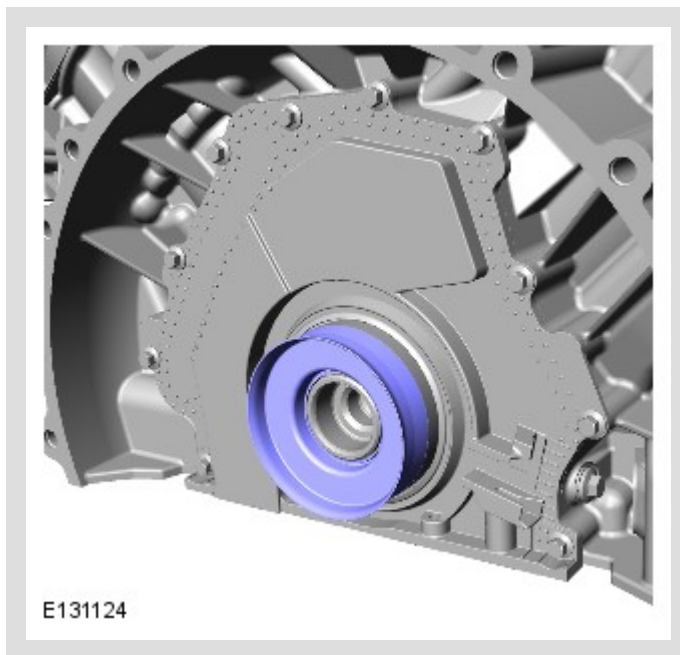
1.

ⓘ CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- A new crankshaft rear seal is supplied with a transit sleeve that must not be removed until the crankshaft rear seal is fully installed.
- Install all the bolts finger tight before final tightening.



Torque: 10 Nm



Remove and discard the transit sleeve.

3.



CAUTIONS:

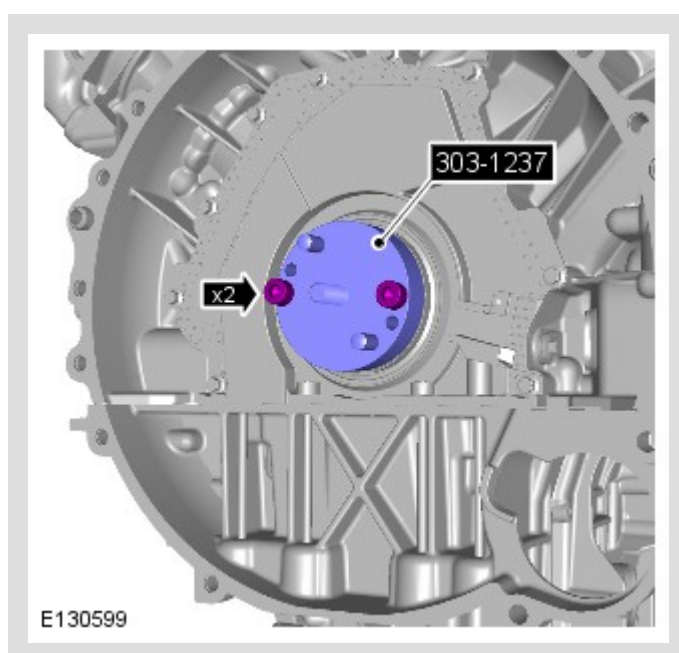
- Make sure that a new component is installed.
- Make sure that the CKP sensor ring is aligned correctly with the special tool pip and that both mating surfaces are fully seated.



Special Tool(s): 303-1237

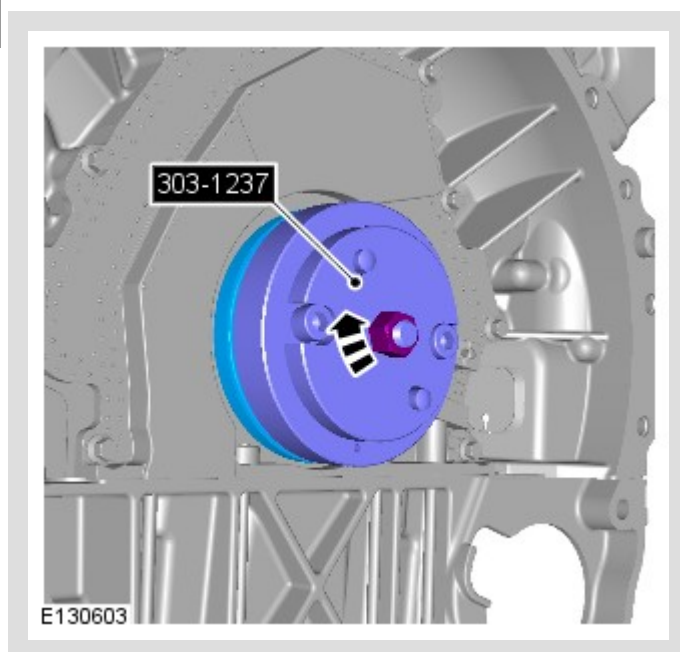
Renew Part: *Rear crankshaft seal* : 1 .

4.

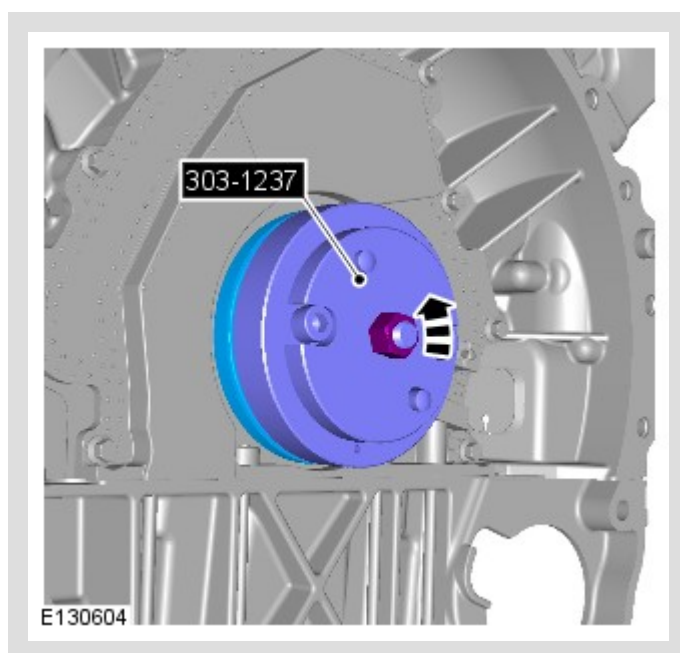


Install the special tool.

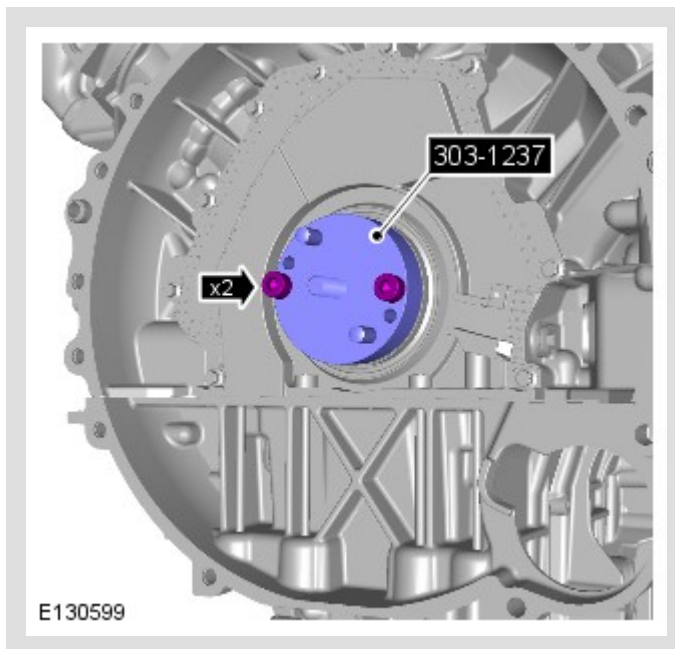
5.



6.



7.

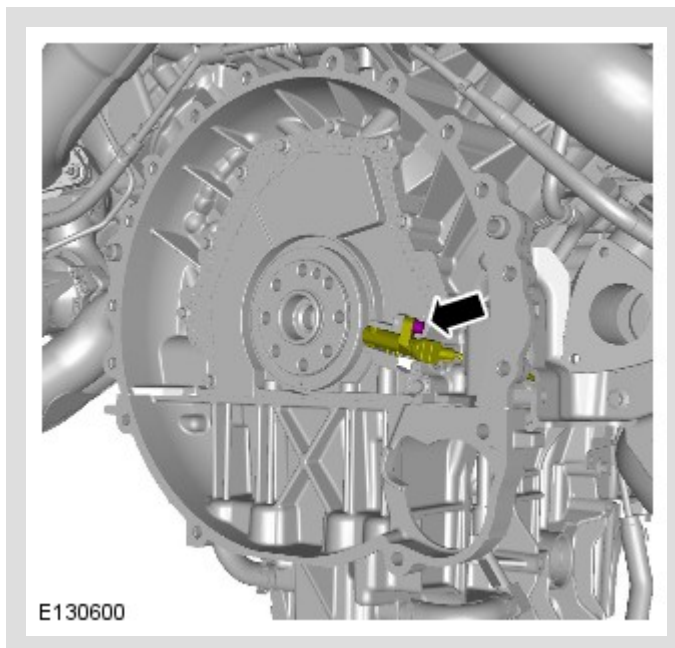


Remove the special tool.

8.

⚠ CAUTION:

Install the CKP sensor correctly into the housing. Failure to follow this instruction may result in damage to the CKP sensor.



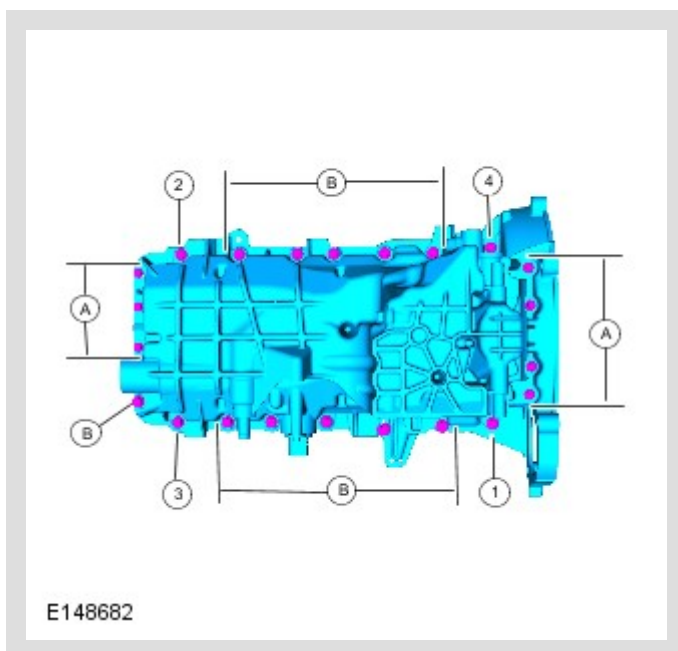
Torque: 5 Nm

⚠ **CAUTIONS:**

- Make sure that the mating faces are clean and free of foreign material.
- Install all the bolts finger tight before final tightening.

⚠ **NOTES:**

- Install a new gasket.
- Tighten the bolts in the indicated sequence.



Renew Part: *Oil pan gasket : 1* .

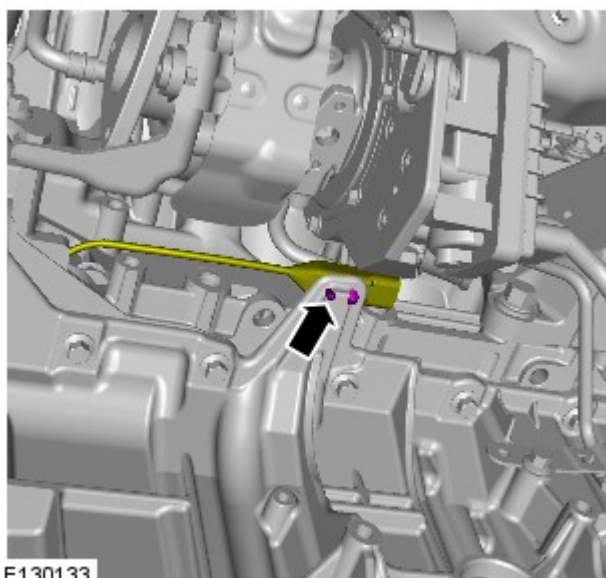
Torque:

1-4 **23 Nm**

B **23 Nm**

A **10 Nm**

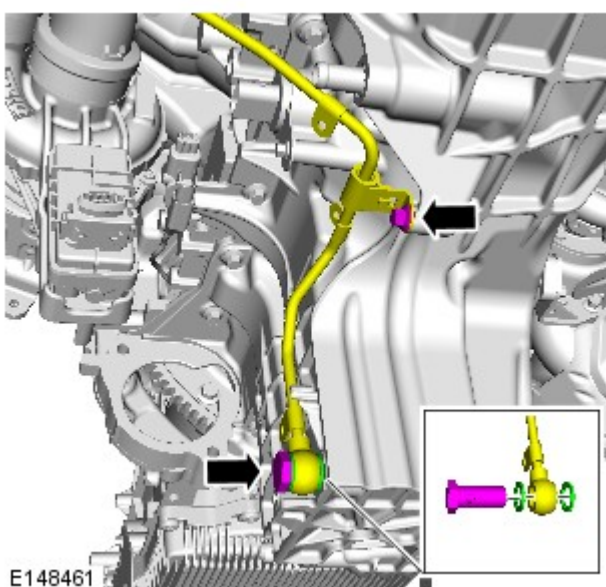
10.



11.

**CAUTION:**

Make sure that the mating faces are clean and free of corrosion and foreign material.

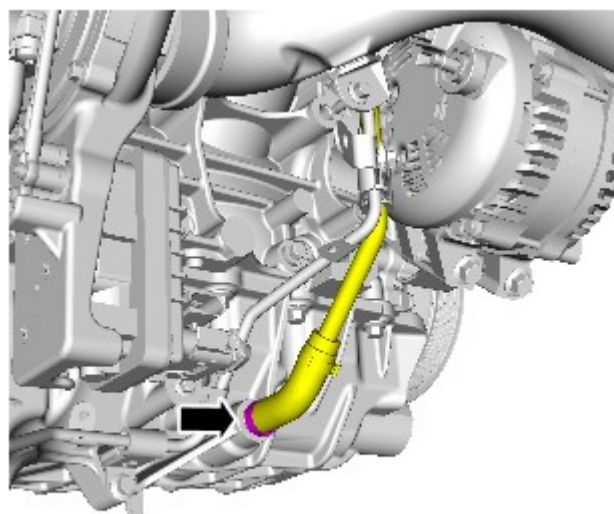


12.



NOTE:

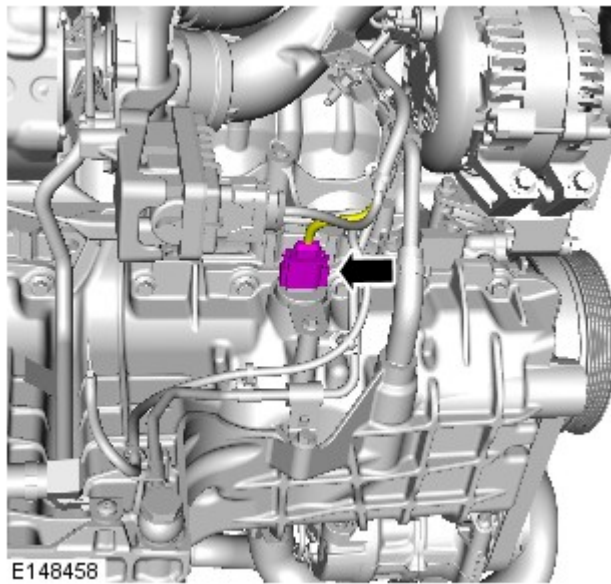
Install a new retaining clip.



E148459

Renew Part: *Retaining clip* : 1 .

13.

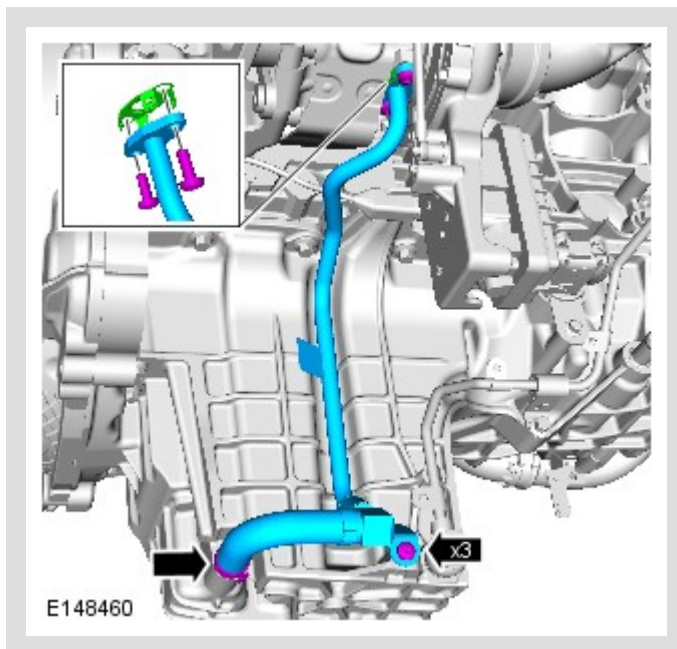


14.



NOTES:

- Remove and discard the blanking caps.
- Install a new gasket.
- Install a new retaining clip.



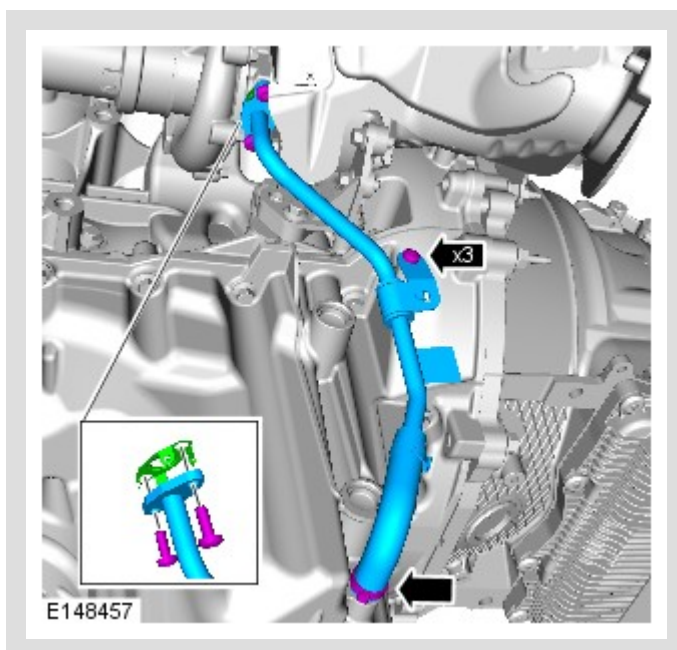
Torque: 10 Nm

15.



NOTES:

- Remove and discard the blanking caps.
- Install a new gasket.
- Install a new retaining clip.

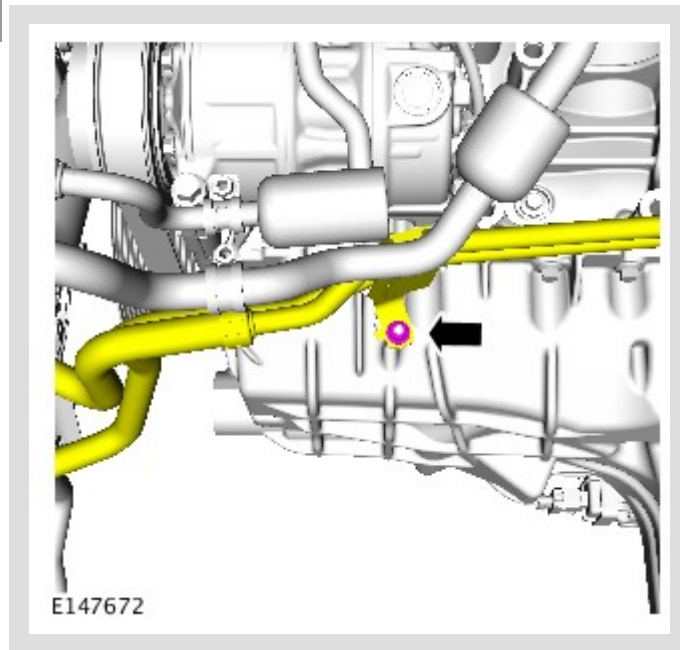


Renew Part: *Turbocharger oil return pipe gasket* : 1 .

Renew Part: *Retaining clip* : 1 .

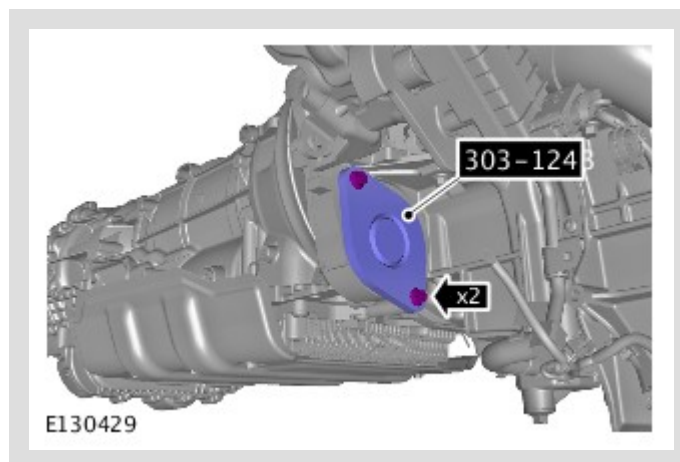
Torque: 10 Nm

16.



Torque: 9 Nm

17.



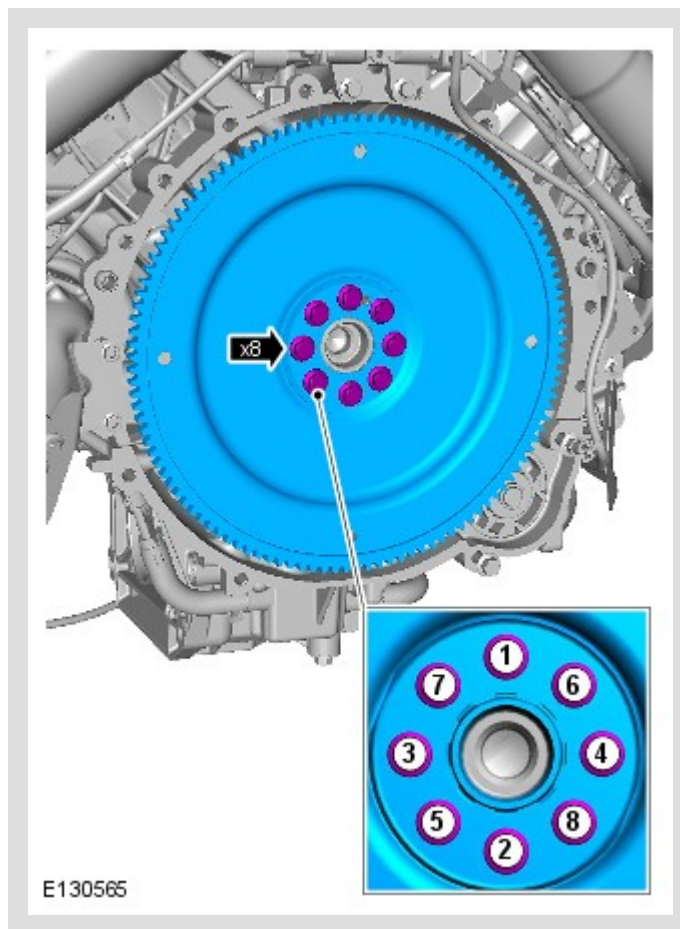
Special Tool(s): 303-1243

18.



CAUTION:

Make sure that new bolts are installed.



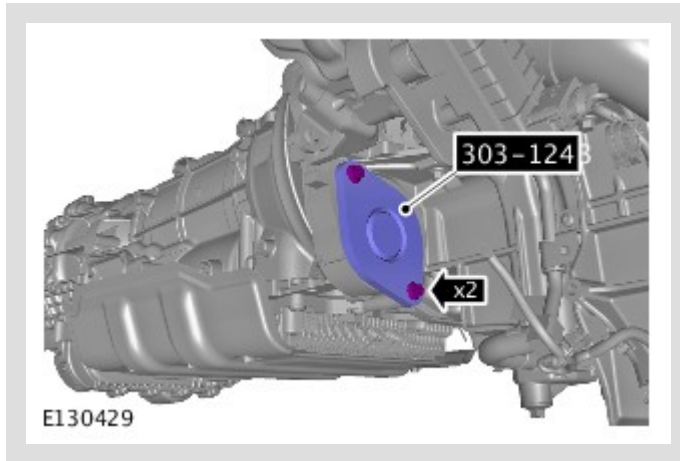
Renew Part: *Flywheel bolts* : 8 .

Torque:

Stage 1 15 Nm

Stage 2 35°

Stage 3 15°



Special Tool(s): [303-1243](#)

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20. Refer to: [Differential Case](#) (205-03 Front Drive Axle/Differential, Removal and Installation).
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21. Refer to: [Transmission - TDV8 4.4L Diesel](#) (307-01B Automatic Transmission /Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal).
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22. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).